

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 June 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Input Management	Enter Country Name (Optional) Enter Life Expectancy Enter Mean Years of Schooling Enter Expected Years of Schooling Enter GNI Per Capita
FR-2	Data Preprocessing	Handle Missing Values Data Cleaning Feature Selection Input Validation
FR-3	HDI Prediction	Load Linear Regression Model Predict HDI Score Calculate Prediction Accuracy
FR-4	Prediction Result Management	Display HDI Score Display HDI Category (Very High / High / Medium / Low) Generate Prediction Summary
FR-5	Data Visualization & Analysis	Generate Heatmaps Generate Scatter Plots Generate Distribution Plots Generate Correlation Matrix Analyze HDI Trends
FR-6	Web Application Interface	Home Page Prediction Page User Input Form Result Display Page
FR-7	Model Management	Train Linear Regression Model Evaluate Model Performance (R^2 Score) Save Model using Pickle (HDI.pkl) Load Saved Model
FR-8	Deployment & Access	Deploy Flask Application Access via Web Browser Provide Real-Time HDI Prediction

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Provide a simple, intuitive, and user-friendly interface for researchers, students, policymakers, and government organizations.
NFR-2	Security	Validate user inputs and ensure secure handling of prediction requests.
NFR-3	Reliability	Produce consistent and accurate HDI predictions using the trained machine learning model.
NFR-4	Performance	Generate prediction results within a few seconds after user input.
NFR-5	Availability	The application should be accessible through standard web browsers at any time.
NFR-6	Scalability	Support multiple users simultaneously and allow future expansion with additional datasets and prediction models.
NFR-7	Maintainability	Enable easy updates to the machine learning model, dataset, and Flask application without affecting existing functionality.
NFR-8	Accuracy	The Linear Regression model should provide high prediction accuracy with a strong R^2 score and minimal prediction error.